

Green infrastructure for sustainable cities

According to UN reports, approximately 55 per cent of the global population lives in urban areas, but this percentage is expected to reach 68 per cent by 2050. Rapid urban growth **surpasses** effective urban and transport planning, **resulting in** worsening air and noise pollution, **urban heat islands**, and limited green spaces. These challenges contribute to significant health risks, including increased **mortality rates** for city **dwellers**.

However, with strategic urban planning and sustainable transport solutions, cities can **mitigate** these harmful effects and create healthier, more liveable environments.

One of the most effective ways to achieve this **is** through active transportation. **Cities** that invest in **pedestrian-friendly** pathways and bike lanes **encourage** physical activity while also minimising **reliance on** cars.

By ensuring safety and convenience, such infrastructure makes walking and cycling more **appealing** to residents, reducing vehicle **congestion** and **emissions**.

Public transport also plays a critical role in **sustainable** urban **mobility**. Expanding metro networks, bus services, and other mass **transit** options can significantly ease traffic **congestion** and lower pollution levels. For example, Paris is enhancing its transit system with new routes to offer **viable** alternatives to car travel, encouraging a shift toward greener **commuting** habits.

Technology is another powerful tool for improving urban transport. Smart traffic **systems** that analyse real-time road conditions **can** **optimise** signal timings, reducing unnecessary **idling** and emissions. Adaptive traffic control strategies **lead to** smoother traffic flow, minimising pollution and enhancing efficiency.

The **transition** to electric vehicles (EVs) **is** a key component of sustainable transport. By investing in EV charging stations, cities make it easier for residents to adopt cleaner transportation options. Many municipalities are prioritising this shift to **align with** long-term **sustainability** goals. Green urban planning extends beyond transport infrastructure. Parks, green roofs, and urban forests help **combat** the heat island effect while providing essential **recreational** spaces. Brisbane, **for instance**, has been actively designing cooler, walkable urban spaces, **setting a precedent for** other cities worldwide. Sustainable road construction materials further contribute to environmental benefits. The **use** of recycled **asphalt** and **permeable** concrete **minimises** resource consumption and improves stormwater management, making roads more **resilient** while reducing their ecological impact.

The **advantages** of sustainable urban transport planning **are far-reaching**. Reduced vehicle usage **leads to** lower carbon emissions and improved air quality. **Access** to green spaces and active transport options **fosters** healthier lifestyles, reducing the risk of lifestyle-related diseases. Efficient public transport and smart traffic systems ease **congestion**, enhancing the daily lives of **commuters**.

Additionally, **investments** in sustainable infrastructure **create** job opportunities and attract responsible businesses, **stimulating** economic growth.

Despite these benefits, several challenges must be **addressed**. Implementing sustainable infrastructure requires significant initial investment, but the long-term **savings** in public health improvements, reduced environmental damage, and economic development **make** it a worthwhile **endeavour**. Strong policies and regulations are essential to **drive** these initiatives forward, ensuring that sustainable urban development remains a priority. Community engagement is also crucial-when residents are actively involved in planning, projects are more likely to **meet** local needs and gain public support. The **journey** toward sustainable cities **is** both necessary and achievable. By prioritising active transport, public transit enhancements, smart traffic solutions, and green infrastructure, urban areas can minimise their environmental **footprint** while improving the quality of life for their residents. While challenges exist, the long-term **benefits** of a low-carbon, people-centric urban environment far **outweigh** the obstacles, **paving the way for** a greener and healthier future.

- Red/blue coloring of words in the sentence indicates subject verb relationship; where 'red' denotes 'subject' and 'blue' denotes 'verb'.

[\[Practice Exercise\]](#)

Vocabulary

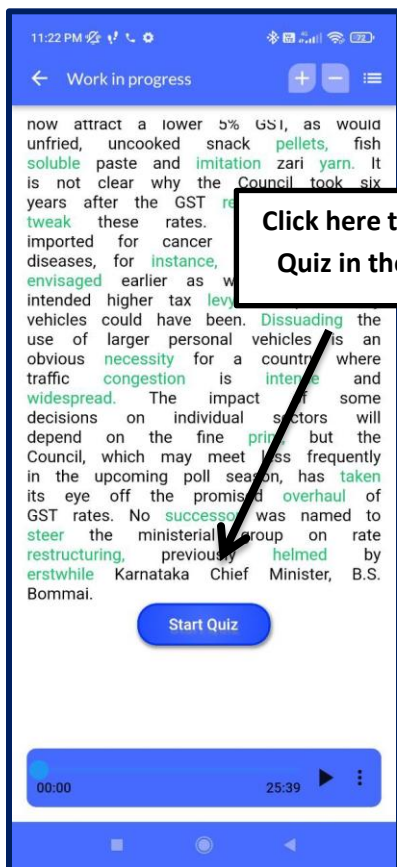
1. **Infrastructure** (noun) – Framework, Facilities, Systems, Structure, Foundation
आधारभूत संरचना
2. **Surpass** (verb) – Exceed, Outdo, Transcend, Outstrip, Excel पार करना
3. **Result in** (phrasal verb) – Cause, Lead to, Bring about, Produce, Generate परिणाम होना
4. **Urban heat island** (noun) – when a city is warmer than the surrounding rural areas. This happens because cities have more buildings, roads, and other structures that absorb and retain heat
5. **Mortality rate** (noun) – Death rate, Fatality rate, Casualty rate, Decease ratio, Lethality मृत्यु दर
6. **Dweller** (noun) – Resident, Inhabitant, Occupant, Citizen, Denizen निवासी
7. **Mitigate** (verb) – Alleviate, Reduce, Lessen, Diminish, Relieve कम करना
8. **Pedestrian** (adjective) – Foot-based, Walker-friendly, Non-motorized, Sidewalk-oriented, Walking पैदल यात्री
9. **Reliance** (on) (noun) – Dependence, Trust, Confidence, Faith, Counting निर्भरता
10. **Appealing** (adjective) – Attractive, Desirable, Inviting, Pleasing, Engaging आकर्षक
11. **Congestion** (noun) – Overcrowding, Jam, Bottleneck, Gridlock, Clog भीड़
12. **Emission** (noun) – Discharge, Release, Ejection, Exhaust, Output उत्सर्जन
13. **Sustainable** (adjective) – Eco-friendly, Renewable, Green, Viable, Maintainable टिकाऊ
14. **Mobility** (noun) – Movement, Transport, Motion, Locomotion, Portability गतिशीलता
15. **Transit** (noun) – Transport, Passage, Conveyance, Movement, Commuting परिवहन
16. **Viable** (adjective) – Feasible, Practical, Workable, Achievable, Sustainable व्यवहार्य
17. **Commute** (verb) – Travel, Shuttle, Journey, Transit, Ride आवागमन करना
18. **Optimise** (verb) – Improve, Enhance, Streamline, Perfect, Maximize अनुकूलित करना
19. **Idle** (verb) – Standby, Loiter, Linger, Wait, Pause निष्क्रिय होना
20. **Lead** (to) (verb) – Result in, Cause, Bring about, Produce, Generate, give rise to वजह बनना
21. **Transition** (noun) – Change, Shift, Conversion, Transformation, Passage परिवर्तन

22. **Align** (with) (verb) – Match, Conform, Coordinate, Harmonize, Adjust संरेखित करना
23. **Sustainability** (noun) – Durability, Endurance, Continuity, Viability, Stability स्थिरता
24. **Combat** (verb) – Fight, Battle, Oppose, Counter, Resist लड़ना
25. **Recreational** (adjective) – Leisure, Entertainment, Amusement, Relaxational, Playful मनोरंजक
26. **For instance** (phrase) – For example, Such as, Like, Namely, As an illustration उदाहरण के लिए
27. **Set a precedent for** (phrase) – Establish a model, Create a standard, Initiate a pattern, Form a benchmark, Start a tradition मिसाल कायम करना
28. **Asphalt** (noun) – Bitumen, Tar, Pavement, Blacktop, Road surface डामर
29. **Permeable** (adjective) – Porous, Penetrable, Absorbent, Spongy, Leaky पारगम्य
30. **Resilient** (adjective) – Tough, Durable, Flexible, Strong, Adaptable मज़बूत
31. **Far-reaching** (adjective) – Extensive, Widespread, Significant, Broad, Expansive दूरगामी
32. **Foster** (verb) – Encourage, Promote, Nurture, Cultivate, Support बढ़ावा देना
33. **Commuter** (noun) – Traveller, Passenger, Daily rider, Transit user, Shuttler आवागमन करने वाला
34. **Stimulate** (verb) – Encourage, Spur, Prompt, Inspire, Activate प्रोत्साहित करना
35. **Address** (verb) – Tackle, Handle, Confront, Deal with, Manage निपटना
36. **Endeavour** (noun) – Effort, Attempt, Venture, Striving, Initiative प्रयास
37. **Drive** (verb) – Power, Propel, Push, Motivate, Force प्रेरित करना
38. **Meet** (verb) – Fulfill, Satisfy, Achieve, Accomplish, Reach पूरा करना
39. **Footprint** (noun) – Impact, Mark, Trace, Imprint, Effect प्रभाव
40. **Outweigh** (verb) – Surpass, Exceed, Eclipse, Overbalance, Prevail अधिक महत्वपूर्ण होना
41. **Pave the way for** (phrase) – Facilitate, Enable, Prepare for, Clear the path for, Open doors for मार्ग प्रशस्त करना

Summary of the Editorial

1. **Rapid Urbanization:** By 2050, 68% of the global population will live in cities, worsening pollution, heat islands, and shrinking green spaces.
2. **Health Risks:** Poor urban planning increases health hazards, including higher mortality rates for city dwellers.
3. **Active Transportation:** Pedestrian-friendly pathways and bike lanes reduce car dependency and promote physical activity.
4. **Public Transport Expansion:** Efficient metro and bus systems (e.g., Paris) can cut traffic congestion and pollution.
5. **Smart Traffic Systems:** Real-time traffic monitoring optimizes signals, reducing emissions and improving flow.
6. **Electric Vehicles (EVs):** Charging infrastructure supports cleaner transport and sustainability goals.
7. **Green Spaces:** Parks, green roofs, and urban forests (e.g., Brisbane) combat heat islands and enhance livability.
8. **Sustainable Road Materials:** Recycled asphalt and permeable concrete reduce environmental impact.
9. **Lower Emissions:** Reduced car usage improves air quality and public health.
10. **Healthier Lifestyles:** Access to green spaces and active transport reduces lifestyle diseases.
11. **Economic Benefits:** Sustainable infrastructure creates jobs and attracts eco-conscious businesses.
12. **High Initial Costs:** Implementation requires investment, but long-term savings justify it.
13. **Policy & Regulations:** Strong governance is needed to prioritize sustainable urban development.
14. **Community Engagement:** Public involvement ensures projects meet local needs and gain support.
15. **Achievable Future:** Sustainable cities are possible through green transport, smart tech, and eco-friendly planning.

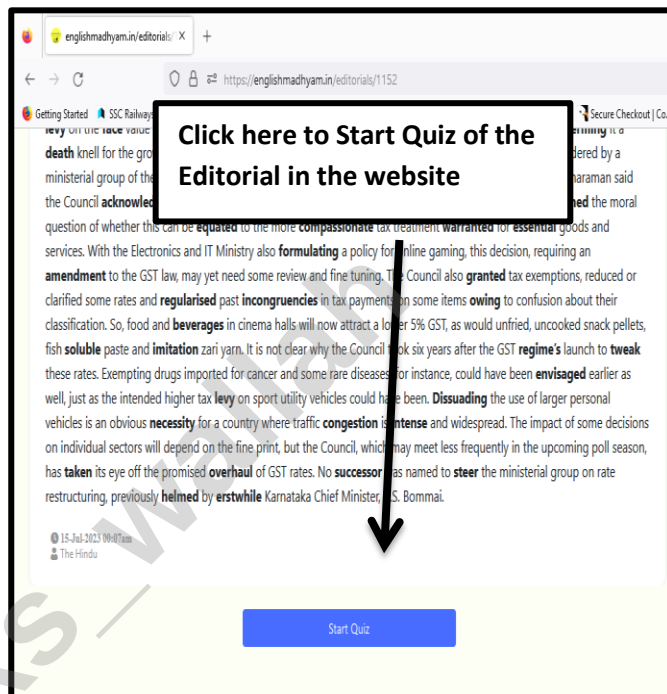
Practice Exercise: SSC Pattern Based



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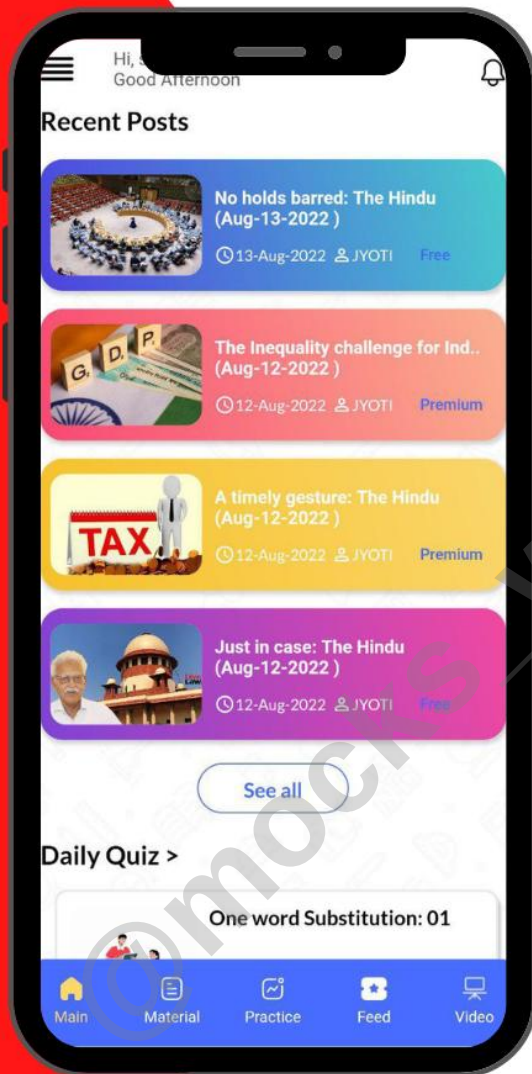
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